

15-2b A Monopoly's Revenue

Consider a town with a single producer of water. [Table 1](#) shows how the monopoly's revenue might depend on the amount of water produced.

Table 1

A Monopoly's Total, Average, and Marginal Revenue

(1)	(2)	(3)	(4)	(5)
Quantity of Water (Q)	Price (P)	Total Revenue ($TR = P \times Q$)	Average Revenue ($AR = TR / Q$)	Marginal Revenue ($MR = \Delta TR / \Delta Q$)
0 gallons	\$11	\$ 0	—	
1	10	10	\$10	\$10
2	9	18	9	8
3	8	24	8	6
4	7	28	7	4
5	6	30	6	2
6	5	30	5	0
7	4	28	4	-2
8	3	24	3	-4

Columns (1) and (2) show the monopolist's demand schedule. If the monopolist produces 1 gallon of water, it can sell that gallon for \$10. If it produces 2 gallons, it must lower the price to \$9 to sell both gallons. If it produces 3 gallons, it must lower the price to \$8. And so

on. If you graphed these two columns of numbers, you would get a typical downward-sloping demand curve.

Column (3) of the table presents the monopolist's *total revenue*. It equals the quantity sold [from column (1)] times the price [from column (2)]. Column (4) computes the firm's *average revenue*, the amount of revenue the firm receives per unit sold. We compute average revenue by taking the number for total revenue in column (3) and dividing it by the quantity of output in column (1). As we discussed in the previous chapter, average revenue always equals the price of the good. This is true for monopolists as well as for competitive firms.

Column (5) of [Table 1](#) computes the firm's *marginal revenue*, the amount of revenue that the firm receives for each additional unit of output. We compute marginal revenue by taking the change in total revenue when output increases by 1 unit. For example, when the firm increases production from 3 to 4 gallons of water, the total revenue it receives increases from \$24 to \$28. Thus, marginal revenue from the sale of the fourth gallon is \$28 minus \$24, or \$4.

[Table 1](#) shows a result that is important for understanding monopoly behavior: *A monopolist's marginal revenue is less than the price of its good*. For example, if the firm raises production of water from 3 to 4 gallons, it increases total revenue by only \$4, even though it sells each gallon for \$7. For a monopoly, marginal revenue is lower than price because a monopoly faces a downward-sloping demand curve. To increase the amount sold, a monopoly firm must lower the price it charges to all customers. Hence, to sell the fourth gallon of water, the monopolist must earn \$1 less revenue for each of the first 3 gallons. This \$3 loss accounts for the difference between the price of the fourth gallon (\$7) and the marginal revenue of that fourth gallon (\$4).

Marginal revenue for monopolies is very different from marginal revenue for competitive firms. When a monopoly increases the amount it sells, there are two effects on total revenue ($P \times Q$):

- *The output effect:* More output is sold, so Q is higher, which increases total revenue.
- *The price effect:* The price falls, so P is lower, which decreases total revenue.

Because a competitive firm can sell all it wants at the market price, there is no price effect. When it increases production by 1 unit, it receives the market price for that unit, and it does not receive any less for the units it was already selling. That is, because the competitive firm is a price taker, its marginal revenue equals the price of its good. By contrast, when a monopoly increases production by 1 unit, it must reduce the price it charges for every unit it sells, and this price cut reduces revenue from the units it was already selling. As a result, a monopoly's marginal revenue is less than its price.

[Figure 3](#) graphs the demand curve and the marginal-revenue curve for a monopoly firm. (Because the firm's price equals its average revenue, the demand curve is also the average-revenue curve.) These two curves always start at the same point on the vertical

axis because the marginal revenue of the first unit sold equals the price of the good. But for the reason we just discussed, the monopolist's marginal revenue on all units after the first is less than the price. Thus, a monopoly's marginal-revenue curve lies below its demand curve.

Figure 3

Demand and Marginal-Revenue Curves for a Monopoly

The demand curve shows how the quantity sold affects the price of the good. The marginal-revenue curve shows how the firm's revenue changes when the quantity increases by 1 unit. Because the price on *all* units sold must fall if the monopoly increases production, marginal revenue is less than the price.

You can see in [Figure 3](#) (as well as in [Table 1](#)) that marginal revenue can even become negative. Marginal revenue is negative when the price effect on revenue outweighs the output effect. In this case, an additional unit of output causes the price to fall by enough that the firm, despite selling more units, receives less revenue.